Alexandre AUTRAN

Mathematics and Machine Learning Student

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Profile

I am a mathematics and machine learning student (*élève normalien*) at ENS Rennes and ENS Paris-Saclay. My research interests include stochastic processes, time-series analysis, quantitative finance, high-dimensional statistics, optimal transport and optimal control. I combine theoretical analysis with numerical methods and machine learning in Python.

Education

2024–2027 **Normalien – ENS Diploma in Mathematics**
Admitted to the ENS through a competitive entrance examination.
Four-year diploma pursued over three years (accelerated track).

2026–2027 **Research Year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, France
Final year of the ENS Diploma (research track, "Prelab" ENS).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France
M2 Fundamental Mathematics. Highest honors (17/20 in the first semester, completed alongside M1).
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 **BSc in Mathematics, minor in Computer Science**, *École Normale Supérieure de Rennes*, France
L3 Fundamental Mathematics. High honors (15.4/20).

2022–2024 **BSc in Mathematics, minor in Computer Science**, *University of Rennes*, France
L2 Mathematics. Highest honors (18/20). Rank: 3/100+.
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

Publications [More details]

[1] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, arXiv preprint, 2026. Submitted to *Stochastic Processes and their Applications*.

[2] **A. Autran**, Conditional Fluctuations of Mean-Field Systems with Stable Common Noise, arXiv preprint, 2026. Under review at *Electronic Journal of Probability*.

[3] **A. Autran** et al., Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles, *Applied Sciences*, 16(11), 5551, 2026.

Research Experience [More details]

Sep 2026–Feb 2027 **Machine Learning Internship**
High-Dimensional Density Clustering Using Local Unimodality.
Supervised by Argyris Kalogeratos (Senior Research Scientist) at Centre Borelli, ENS Paris-Saclay, France.

Jun–Aug 2026 **M1 Research Internship**
Optical Modeling of VO_2-Based Thermochromic Oxide Nanocomposites.
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the University of Liège, Belgium.

Apr–Jun 2026 **Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Jun–Aug 2025 **L3 Research Internship**
Lattice-Based Post-Quantum Cryptography.
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevic (Research Scientist) at Inria, France.

May–Jun 2025 **Research Internship**
Algebraic Counting of the Real Roots of Polynomials.
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

GitHub Projects [More details]

2026 **Time-Series Statistical Arbitrage under Transaction Costs and Forecast Uncertainty**
Built and tested a statistical arbitrage strategy on six cryptocurrency perpetual futures, combining return forecasting, cost-aware portfolio optimization, and walk-forward validation with bootstrap inference.

2026 **Comparing Dense and Sparse Neural Models for Constrained Graph Generation**
PyTorch implementation of diffusion-based graph generation with hard connectivity and edge-budget guarantees, with a focus on structural validity and efficient generation.

Other Activities

2024–2025 **ENS Oral Examinations Preparation**
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).

2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, and for first- and second-year BSc students at the University of Rennes.

2023–2024 **BSc Student Representative**
Representative on the improvement council of the Mathematics Department of the University of Rennes.

Before 2022 **High School Competitions**
Participation in the Concours Général in Mathematics, Physics, and Philosophy, and the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

Technical Skills

Stochastic Modeling: Stochastic processes, mean-field systems, fractional Brownian motion.
Statistics: Statistical inference, time-series forecasting, walk-forward validation, bootstrap methods.
Machine Learning: PyTorch, scikit-learn, ridge regression, gradient boosting, diffusion models.
Optimization: CVXPY, convex optimization, robust portfolio allocation, optimal control.
Scientific Computing: Python (NumPy, pandas), MATLAB, R.
Programming & Tools: Python, C/C++, Java, SQL, Git, LaTeX.

Languages

French: native. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: fantasy and science fiction literature, chess.

Academic References

François Bolley — francois.bolley@ens-rennes.fr
Full Professor, IRMAR, ENS de Rennes.
Director of the Mathematics Department at ENS de Rennes.

Jean-Christophe Breton — jean-christophe.breton@univ-rennes.fr
Full Professor, IRMAR, University of Rennes.

Françoise Dal'Bo — francoise.dalbo@univ-rennes.fr
Full Professor, IRMAR, University of Rennes.
Director of the International Research Laboratory IFUMI.

Martin Gander — martin.gander@unige.ch
Full Professor, Section of Mathematics, University of Geneva.

Argyris Kalogeratos — argyris.kalogeratos@ens-paris-saclay.fr
Senior Research Scientist, Centre Borelli, ENS Paris-Saclay.